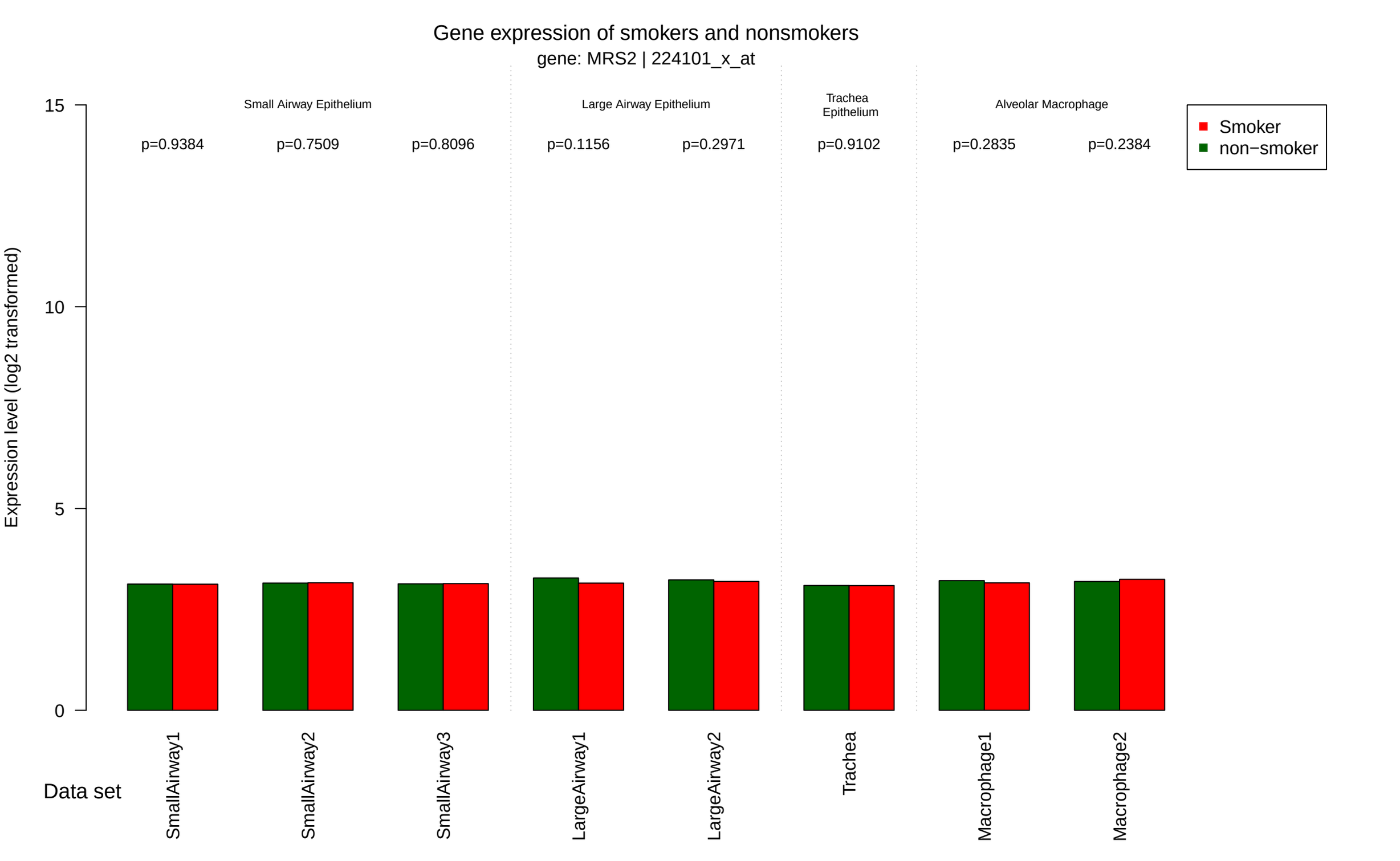
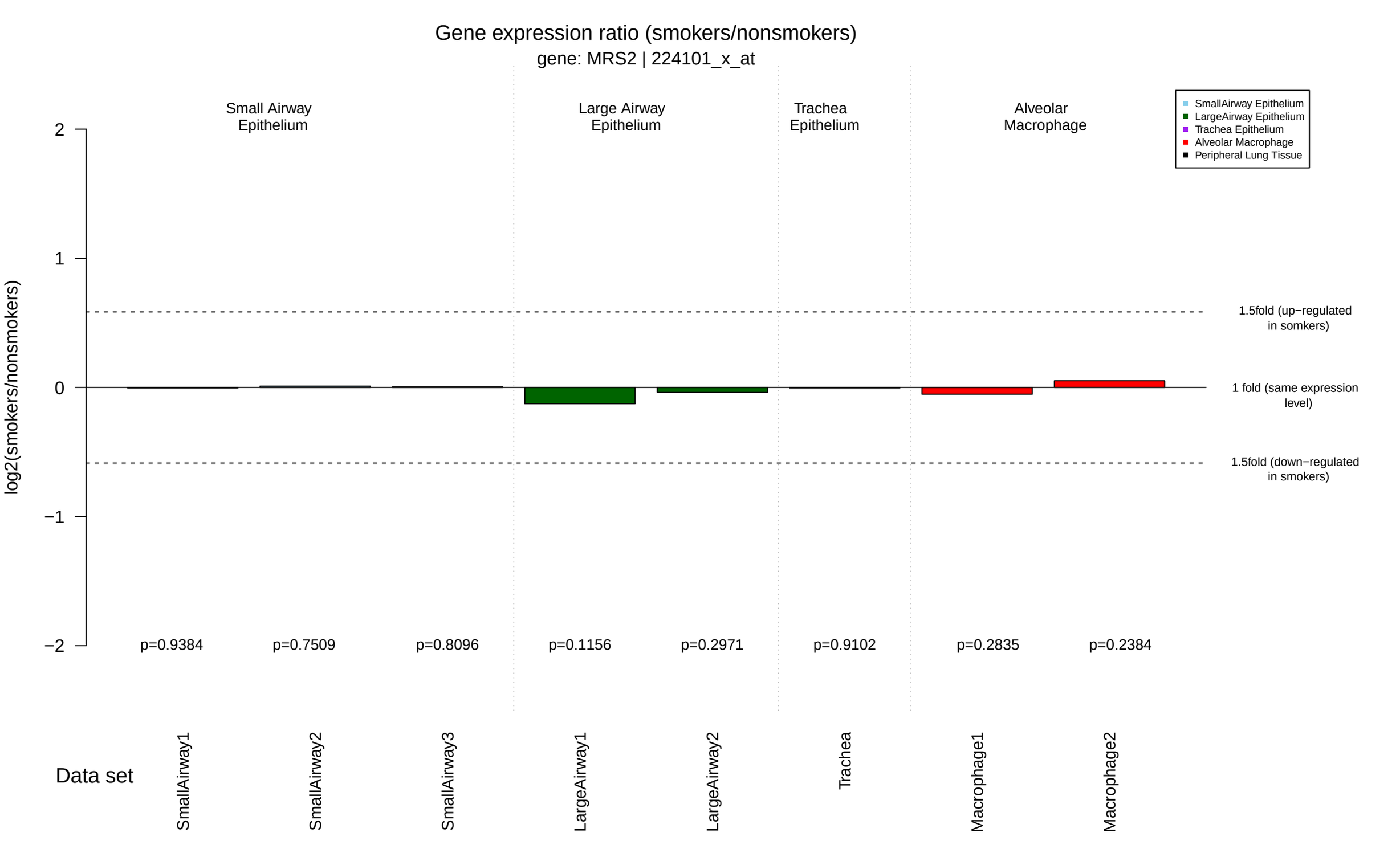
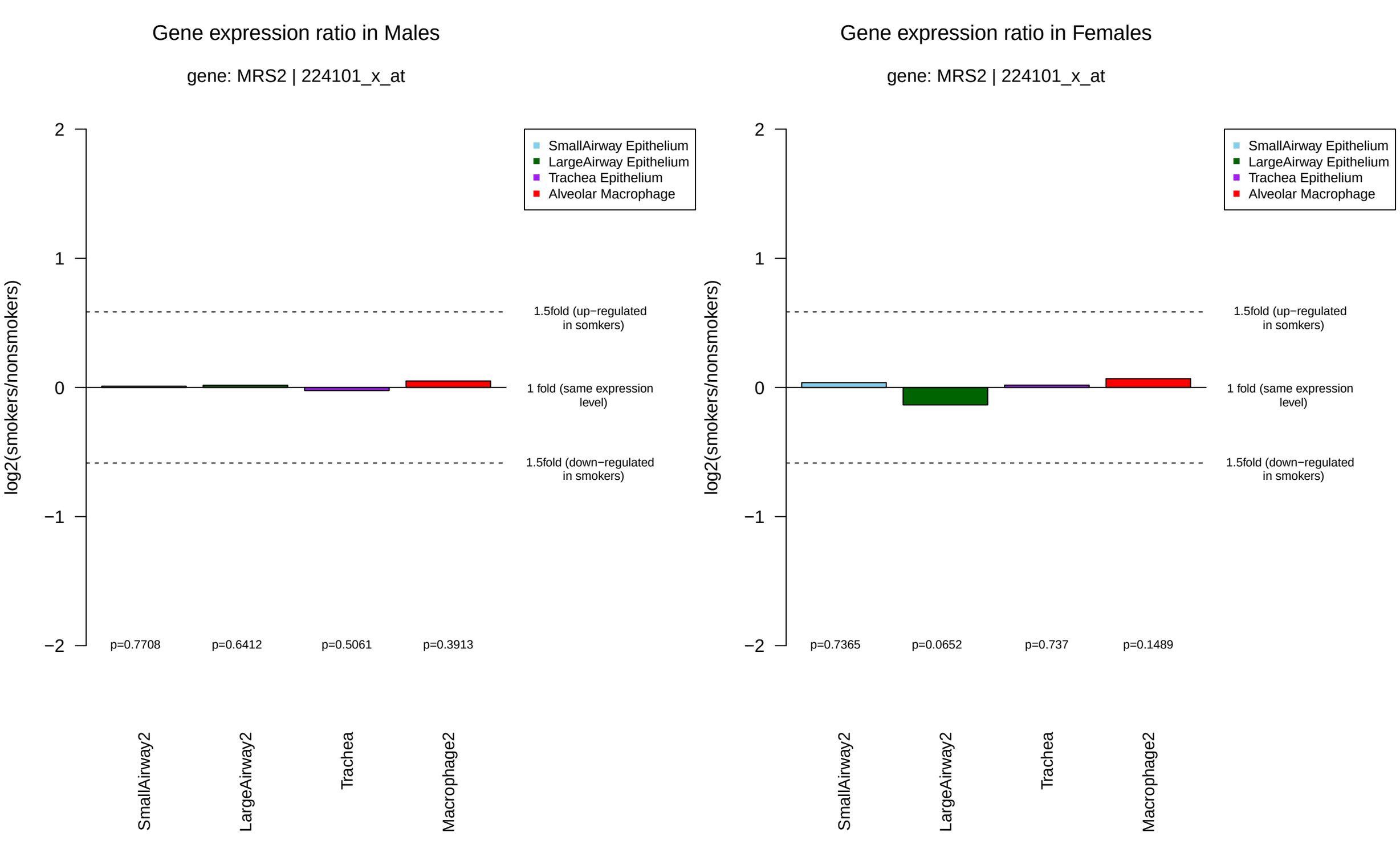
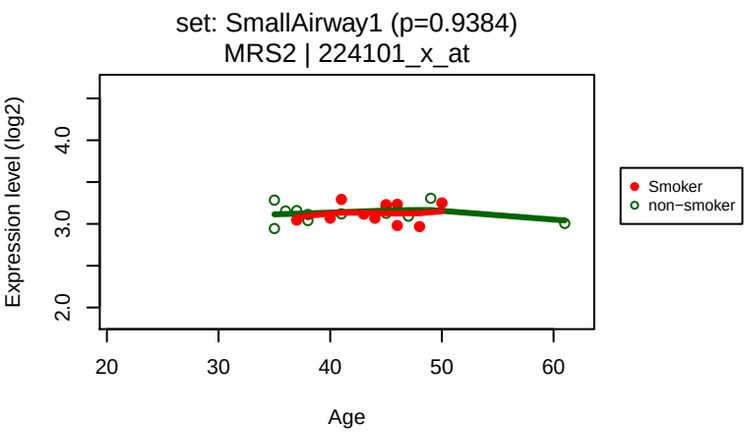
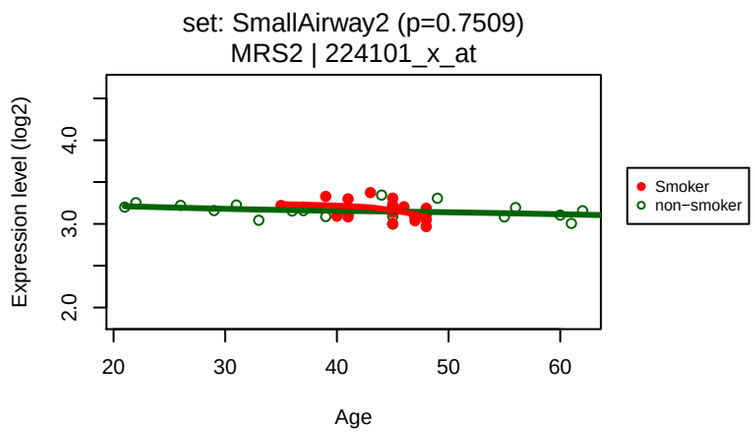
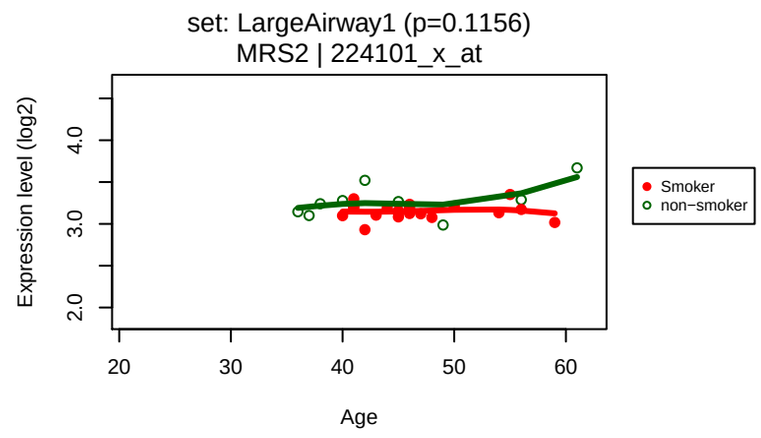

Gene expression of smokers and nonsmokers
gene: MRS2 | 224101_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9384
p=0.7509
p=0.8096
p=0.1156
p=0.2971
p=0.9102
p=0.2835
p=0.2384
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

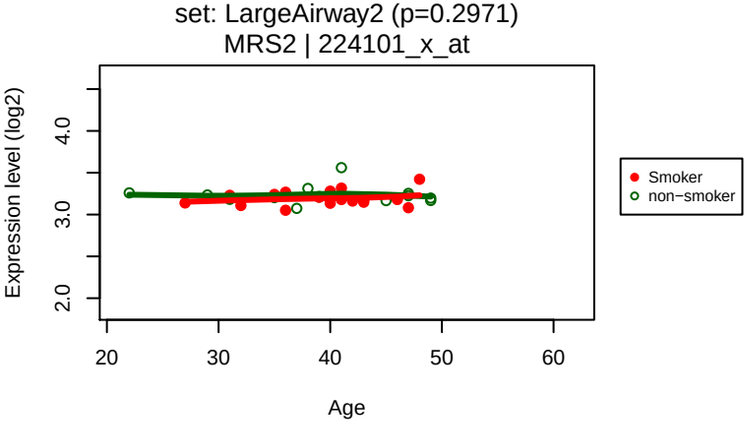

Gene expression ratio (smokers/nonsmokers)
gene: MRS2 | 224101_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9384
p=0.7509
p=0.8096
p=0.1156
p=0.2971
p=0.9102
p=0.2835
p=0.2384
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

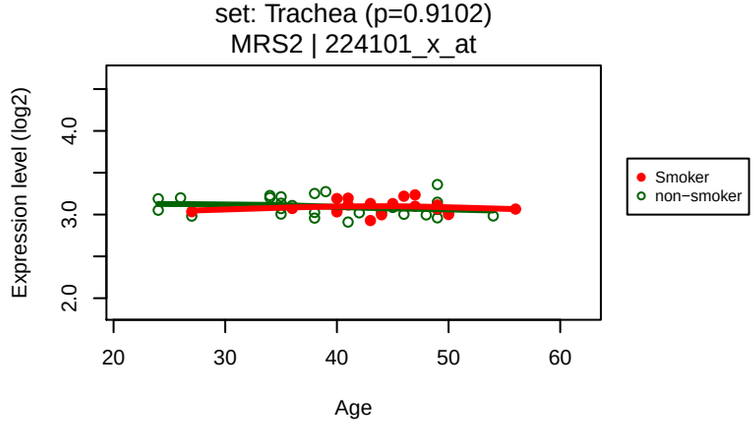

Gene expression ratio in Males
gene: MRS2 | 224101_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7708
p=0.6412
p=0.5061
p=0.3913
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MRS2 | 224101_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7365
p=0.0652
p=0.737
p=0.1489
SmallAirway2
LargeAirway2
Trachea
Macrophage2

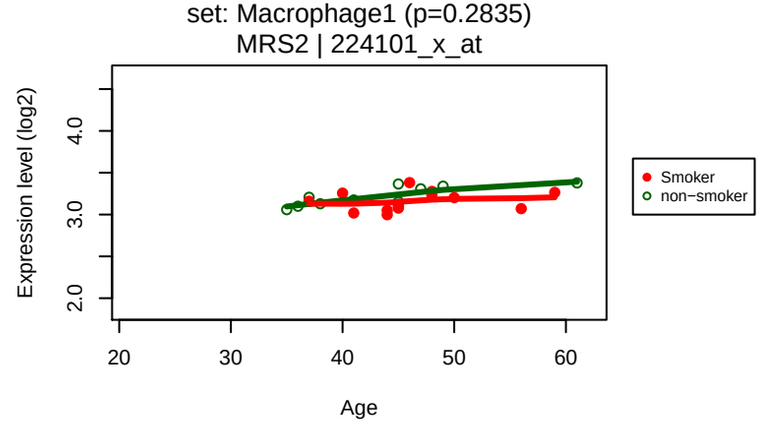

set: SmallAirway1 (p=0.9384)
MRS2 | 224101_x_at
Expression level (log2)
Age
Smoker
non−smoker

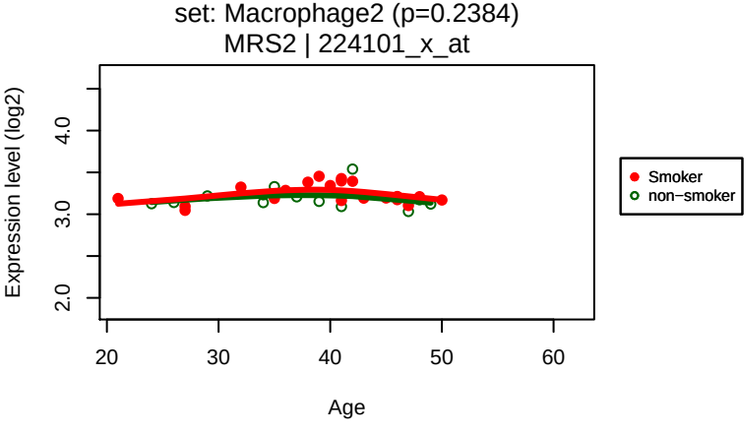

set: SmallAirway2 (p=0.7509)
MRS2 | 224101_x_at
Expression level (log2)
Age
Smoker
non−smoker

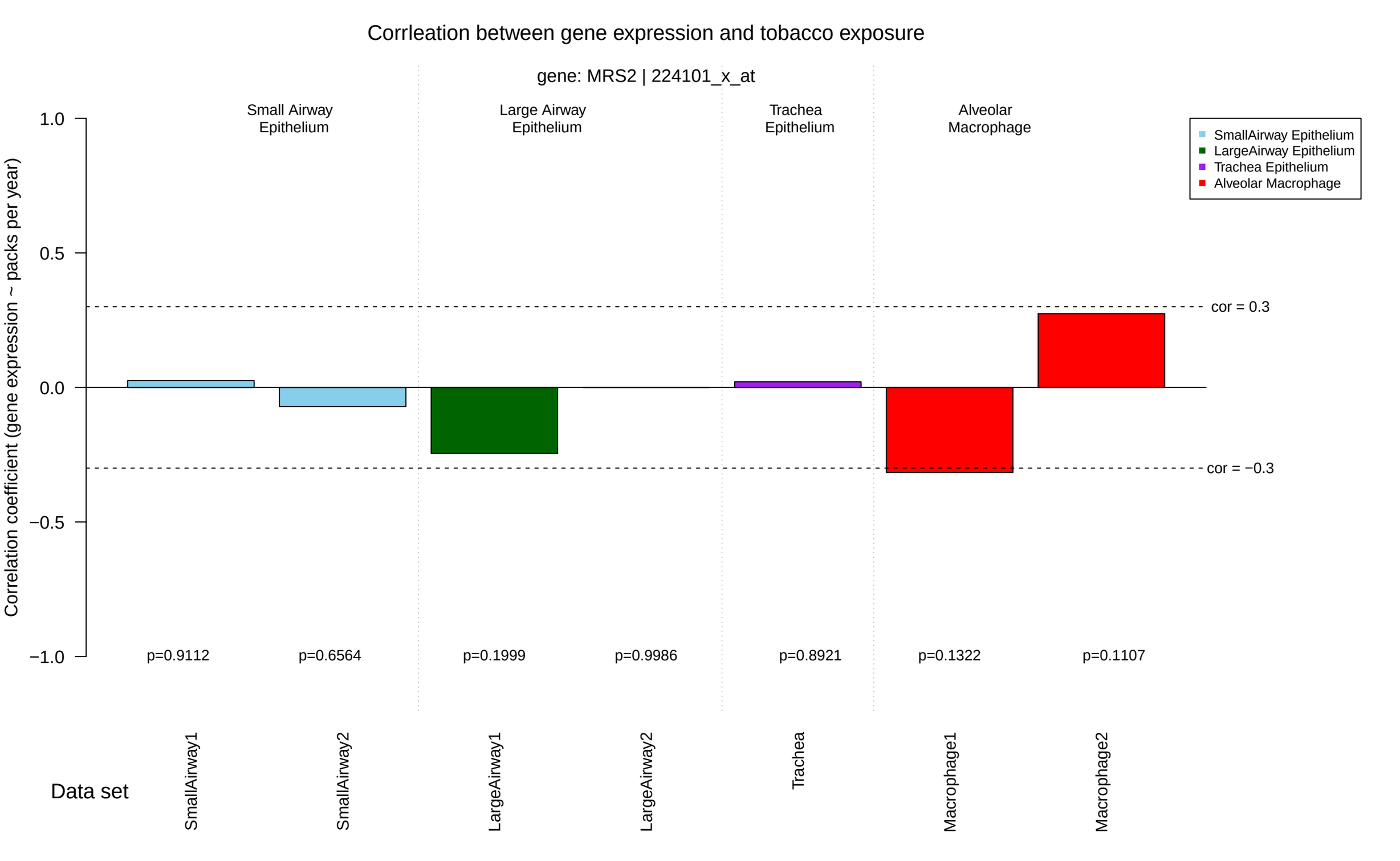

set: LargeAirway1 (p=0.1156)
MRS2 | 224101_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2971)
MRS2 | 224101_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9102)
MRS2 | 224101_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2835)
MRS2 | 224101_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2384)
MRS2 | 224101_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MRS2 | 224101_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9112
p=0.6564
p=0.1999
p=0.9986
p=0.8921
p=0.1322
p=0.1107
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2